S5 — Glossary

Operational terms and evidence boundaries

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Abstract Definitions for the operational constructs used in the ds004587 EAP analysis scaffold. Definitions distinguish implemented measurements, permitted interpretations, and claims not established by this observational design.

Reading rule. An **implemented** entry describes a versioned calculation. An **interpretation** states its narrow empirical reading. A **not established** entry identifies inferences the present study cannot support.

A – D

Anchor feature — A physiology-derived measured variable aligned to the neural analysis grid, such as an ECG, PPG, or respiratory summary. **Not established:** a direct measure of EAP strength, interoceptive awareness, or causal control of neural dynamics.

Association — A statistical relation between measured variables. In this study, an association can be within participant or between participant and must state its time scale and correction family. **Not established:** temporal precedence or causality.

BH-FDR — Benjamini–Hochberg false-discovery-rate correction, used within explicitly declared endpoint families. A threshold of $q < 0.05$ controls the expected false-discovery proportion under its assumptions. (Benjamini & Hochberg, 1995)

BIDS provenance — The traceable relationship between an analysis row and its source participant, session, run, event, behavioral record, and data-release version.

Condition contrast — A comparison between rest and illusion-perception summaries. It is not an illusion-specific neural mechanism because the conditions differ in visual input, eye state, engagement, and response demands.

E – L

EAP (Embodied Anchoring Principle) — A speculative NDT extension proposing that cardiac and respiratory state may constrain neural dynamics. **Implemented here:** only candidate measured physiological proxies and conditional association tests. **Not established:** cycle-level entrainment, causal gating, or a unique physiological-to-neural pathway.

Epoch — A defined time interval of a continuous recording. Its duration, overlap, and condition assignment must be frozen in the run manifest before inference.

Geometry contract — The run-level validity metadata that states whether a specified neural-geometry output, especially a Jacobian, meets the pipeline's operational requirements. A failed contract makes that endpoint unavailable, not null.

HRV (heart-rate variability) — Variation in NN intervals derived from ECG R peaks. Its interpretation depends on R-peak quality, artifact handling, integration window, and stationarity. (Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996)

Illusion strength — A dataset-defined stimulus parameter intended to manipulate the contextual bias of a visual illusion. An association with this label is not proof of subjective illusion sensitivity unless the behavioral mapping is verified.

M – R

MNDM — The versioned computational pipeline used to export operational neural-coordinate, local-flow, and optional physiological surfaces. Its output schema and configuration are part of the measurement definition. (Langell, 2026)

MNJ (Meta-Noetic Jacobian) — The estimated local Jacobian of an operational neural flow field. A common magnitude summary is $\|J\|_F = \sqrt{\sum_{i,j} J_{ij}^2}$. **Interpretation:** a model-dependent local deformation summary. **Not established:** direct physical force, stability, or cognitive efficiency.

MNPS coordinate — One component of the exported three-coordinate neural chart, conventionally (m, d, e). **Interpretation:** a versioned chart coordinate. **Not established:** a directly observed latent psychological dimension.

Null result — An estimated effect that does not meet a declared decision threshold. It does not demonstrate equality or absence of all plausible effects; precision and confidence intervals are required.

PPG (photoplethysmography) — An optical peripheral pulse signal. Quality, sensor placement, and its relation to ECG must be documented before it is used as an anchor feature.

Respiratory phase — Instantaneous position within a respiratory cycle. It requires an explicit extraction method and validated timing alignment; it cannot be inferred from a session-level respiration-rate summary.

S – Z

Synchronization gate — A prespecified decision requiring adequate evidence for EEG–physiology timing alignment before event-, phase-, lag-, or direction-sensitive analyses are run.

Variance schedule — In the EAP/NDT theoretical notation, $\sigma(t)$ controls stochastic variability in a state-dynamics equation. **Not established here:** an observed or causally identified biological variance schedule.

Wilcoxon signed-rank test — A nonparametric paired test used for the planned participant-level rest–illusion contrast. It tests a paired distributional shift under its assumptions and is reported with an effect size and interval. (Wilcoxon, 1945)

Bibliography

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Langell, R. (2026,). *NeuralManifoldDynamics: a versioned measurement contract for low-dimensional neural-manifold trajectories*. https://doi.org/10.5281/zenodo.20305778

Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. (1996). Heart rate variability: standards of measurement, physiological interpretation, and clinical use. *Circulation*, *93*(5), 1043–1065. https://doi.org/10.1161/01.CIR.93.5.1043

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, *1*(6), 80–83. https://doi.org/10.2307/3001968